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Freedom of choice boosts midfrontal theta power during affective feedback processing of goal-directed actions

Maren Giersiepen [*,1], Simone Schütz-Bosbach, Jakob Kaiser

Ludwig-Maximilians-University, General and Experimental Psychology, Leopoldstr. 13, D-80802 Munich, Germany

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ABSTRACT

Sense of agency, the feeling of being in control of one's actions and their effects, is particularly relevant during goal-directed actions. During feedback learning, action effects provide information about the best course of action to reinforce positive and prevent negative outcomes. However, it is unclear whether agency experience selectively affects the processing of negative or positive feedback during the performance of goal-directed actions. As an important marker of feedback processing, we examined agency-related changes in midfrontal oscillatory activity in response to performance feedback using electroencephalography. Thirty-three participants completed a reinforcement learning task during which they received positive (monetary gain) or negative (monetary loss) feedback following item choices made either by themselves (free-choice) or by the computer (forced-choice). Independent of choice context, midfrontal theta activity was more enhanced for negative than positive feedback. In addition, free, compared to forced choices increased midfrontal theta power for both gain and loss feedback. These results indicate that freedom of choice in a motivationally salient learning task leads to a general enhancement in the processing of affective action outcomes. Our findings contribute to an understanding of the neuronal mechanisms underlying agency-related changes during action regulation and indicate midfrontal theta activity as a neurophysiological marker important for the monitoring of affective action outcomes, irrespective of feedback valence.

1. Introduction

Sense of agency, the feeling of being in control of one's actions and their perceivable results in the environment (Gallagher, 2000; Haggard, 2017; Haggard & Eitam, 2015), is crucial in constructing a sense of self (Gentsch & Schütz-Bosbach, 2015). Contingent upon situational demands, moment-to-moment fluctuations in agency experience continuously affect an individuals' self-representation (Gallagher, 2013; Haggard, 2017; Hommel, 2015; Verschoor & Hommel, 2017). To manipulate the sense of agency, many studies have asked participants to either freely choose from different response options or forced them to select a predetermined option (e.g., Barlas et al., 2017; Barlas & Kopp, 2018; Barlas & Obhi, 2013; Hassall et al., 2019; Murayama et al., 2015; Zheng et al., 2020). The current study extends previous work by investigating how freedom of choice influences the neural processing of performance feedback in a motivationally salient learning context.

Understanding the functional role of freedom of choice in the emergence of agency experience is most relevant in situations where agents need to effectively interact with their environment to accomplish their goals (Leotti et al., 2015; Ly et al., 2019; Murayama et al., 2015). When trying to reach a goal, affective feedback in the form of positive and negative action effects, signals whether behavioral adjustments are required to attain desired outcomes and avert undesirable ones in the future. An agents' belief in the ability to effectively translate the presented feedback into subsequent action adaptation thus likely contributes to its utilization during goal-directed actions (Moscarello & Hartley, 2017).

Functional magnetic resonance imaging (fMRI) studies suggest that the anterior cingulate cortex (ACC) facilitates the evaluation of affective feedback (e.g., Bush et al., 2002; Calabro et al., 2023; Luft, 2014; Marco-Pallarés et al., 2007). These findings are corroborated by electroencephalographic (EEG) recordings that provide a direct measure of brain activity with high temporal precision (Foti et al., 2015; Kaiser et al., 2021; Luft, 2014). To date, most EEG studies have focused on event-related potentials (ERPs) to assess the neuronal mechanisms of affective feedback processing. Among them, the feedback-related

* Corresponding author.
E-mail addresses: m.giersiepen@psy.lmu.de (M. Giersiepen), s.schuetzbosbach@psy.lmu.de (S. Schütz-Bosbach), j.kaiser@psy.lmu.de (J. Kaiser).
[1] URL: https://www.psy.uni-muenchen.de/expneurocog/people/lab-members/maren_giersiepen/index.html

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negativity (FRN) is the most frequently examined ERP (also referred to as Reward Positivity; Luft, 2014; Proudfit, 2015). The FRN is a negative going component with higher amplitude for negative than positive feedback, commonly peaking 250–350 ms after feedback onset (see Luft, 2014). Consistent with its maximal deflection at midfrontal scalp sites, the ACC has been proposed as the neural generator of the FRN (Hauser et al., 2014). Importantly, some studies indicate an increase in FRN amplitude for self-, rather than externally determined action outcomes during reinforcement learning, indicating an influence of agency on feedback processing in the ERP domain (e.g., Bellebaum et al., 2010; Weismüller et al., 2019; Yeung et al., 2005).

While ERPs provide valuable insights into the influence of agency on affective feedback processing, recent research has highlighted the crucial role of oscillatory activity for feedback processing (Beste et al., 2023; Cavanagh & Frank, 2014; Luft, 2014). More specifically, midfrontal oscillations in the theta range (4–7 Hz) are believed to represent a central mechanism for the updating of task-relevant information, indicated by an increase in oscillatory power in response to performance feedback (e.g., Cavanagh et al., 2010; Kaiser et al., 2021; Luft et al., 2013; Weismüller et al., 2019). During feedback presentation, negative action outcomes typically evoke larger theta power than positive outcomes (e.g., Kaiser et al., 2021; Luft et al., 2013). This power increase for negative outcomes has been suggested to partially reflect cognitive control processes associated with the ACC (Cavanagh & Frank, 2014; Holroyd & Umemoto, 2016; Kaiser et al., 2019; Luft et al., 2013; Weismüller et al., 2019; Zheng et al., 2020). In addition, several studies report a relation of feedback-induced midfrontal theta power and learning success, with mixed findings whether this association is driven only by negative (Luft et al., 2013; Van de Vijver et al., 2011) or both negative and positive feedback (Kaiser et al., 2021).

Given that agency experience enhances our ability to effectively reach our goals (Cordova & Lepper, 1996; Eitam et al., 2013; Murayama et al., 2015), as well as the importance of midfrontal theta oscillations for the processing of affective feedback, it is reasonable to assume that activity in this frequency range is sensitive to the experience of agency during goal-directed actions. Thus, changes in agency experience are expected to modulate feedback-driven midfrontal theta activity. Yet, most previous studies investigated how sense of agency influences the processing of arbitrary, neutral action outcomes, for example, by letting participants actively produce or passively perceive sensory action effects without any rewarding or punishing value (e.g., Gentsch & Synofzik, 2014; Haggard & Eitam, 2015; Kaiser et al., 2021). In these studies, actively produced, compared to passively perceived action effects typically elicit decreased neural reactivity (Brown et al., 2013). This attenuation effect is assumed to result from the neural cancellation of sensory feedback that can be predicted for self- but not externally induced action effects (Gentsch & Schütz-Bosbach, 2015). However, neutral, in contrast to instrumental affective outcomes, carry no rewarding or punishing value and may thus be less relevant for subsequent action selection. Therefore, results from studies employing non-affective action outcomes leave open how the sense of agency influences the processing of affective feedback in motivationally salient contexts.

To study how freedom of choice and associated agency experience influence feedback processing during goal-directed actions, experimental tasks should ensure that action outcomes carry positive and negative affective value and implement reward schemes that provide agents with the possibility to learn item-outcome associations to adapt their behavior accordingly (Eitam et al., 2013; Moscarello & Hartley, 2017). However, most studies investigating the influence of sense of agency on affective feedback processing employed randomized reward schemes at chance level, with half of the trials providing positive and negative feedback, respectively (e.g., Gentsch et al., 2015; Hassall et al., 2019; Mei, Yi, et al., 2018; Mühlberger et al., 2017; Yeung et al., 2005; Zheng et al., 2020). This offers the advantage of ensuring an equal number of trials with positive and negative feedback, which enables us to consider expectancy effects that might influence the processing of affective action outcomes. However, when actions result in reward or loss feedback solely by chance, the instrumental value of action outcomes decreases. This renders performance feedback uninformative for selecting the best course of action, thereby potentially reducing the perceived relevance of action outcomes in these studies. As feedback-evoked midfrontal theta activity seems to be particularly relevant during goal-directed actions (Cavanagh et al., 2010), the neural processing of affective feedback conceivably differs, if action outcomes can be used to determine the best course of action, compared to when not.

From the studies focusing on agency-related changes in affective feedback processing, only a few have specifically investigated valence-specific effects. These studies aimed to determine whether the sense of agency primarily impacts the processing of negative or positive performance feedback. However, among the studies that explored this, inconsistent effects were found (see Kaiser et al., 2021). For midfrontal theta activity, two studies report an increased impact of feedback, operationalized as monetary gains and losses, for active compared to passive choices (Weismüller et al., 2019; Zheng et al., 2020). Whereas the effect of feedback valence did not differ for active and passive choices in one study (Weismüller et al., 2019), the other study found a larger impact of choice on theta power for negative, compared to positive feedback (Zheng et al., 2020). Thus, it is currently unclear, whether freedom of choice leads to a general enhancement, or a valence-specific bias in the processing of affective feedback. A selective increase in midfrontal theta power for either positive or negative feedback could be seen as evidence that sense of agency induces an affective processing bias. For example, some studies suggested that humans feel more agency for positive than negative action outcomes, suggesting a self-serving processing bias for positive feedback (Forsyth, 2008; Gentsch et al., 2015; Kaiser et al., 2021).

Building upon previous work, the current study aimed to clarify how the sense of agency, manipulated by varying freedom of choice in a within-subjects design, influences the processing of affective feedback in a motivationally salient context. Using EEG, we measured neural reactivity to positive and negative feedback while participants completed a reinforcement learning task. Importantly, whereas in most previous studies, reward and punishment was determined by chance alone, we implemented a reward schedule that allowed participants to maximize their rewards through learning. By granting a performance-dependent monetary bonus at the end of the experiment, we additionally warranted the affective value of action outcomes. Building on previous work (Cavanagh & Frank, 2014; Luft et al., 2013; Weismüller et al., 2019; Zheng et al., 2020), we expected a larger increase in theta power for monetary losses compared to gains. In addition, as performance feedback was informative for subsequent action selection during free-, but not during forced-choice blocks, we expected an increase in feedback-induced midfrontal theta activity, if participants were endowed with the freedom to choose between different response options (Weismüller et al., 2019; Zheng et al., 2020). Crucially, we also examined whether the effect of choice differed for positive and negative feedback. We thereby aimed to clarify whether freedom of choice induces a valence-specific processing bias, meaning a selective increase in the processing of either positive or negative feedback, or a general enhancement in the processing of affective action outcomes. We complemented our neurophysiological measure on agency-related differences in feedback processing with explicit agency ratings. Participants were asked to indicate whether they felt they had freedom to choose their actions or influence the amount of money they gained. To conclude, using explicit agency judgments and EEG to measure the neural activity during the presentation of affective feedback following free and forced choices in a reinforcement learning task, we aimed to elucidate the role of sense of agency during goal-directed actions.

2. Methods

2.1. Participants

Eligibility criteria for participation included English language proficiency, an age between 18 and 45 years, absence of acute neurological or psychiatric disorders, no current intake of medications affecting neural functioning, and normal or corrected-to-normal visual acuity without color vision deficiencies. Thirty-three participants (22 female, 11 male, 30 right-handed) participated in the study. After terminating data collection, one participant was excluded due to an exceedingly high error rate across free- and forced-choice blocks (i.e., > 2.5 standard deviations from the mean; 51.99%). Two additional participants were excluded after pre-processing, owing to noise artifacts that resulted in less than 50 trials for the main analysis in at least one experimental condition (see Data preprocessing). The final test set thus consisted of 30 participants (20 female, 10 male, 27 right-handed) with a mean age of 24.8 years ($SD = 4.1$), the sample size being in line with previous studies in this field (e.g., Kaiser et al., 2021; Luft et al., 2013; Weismüller et al., 2019; Zheng et al., 2020). Participants were compensated with participation credits or financial reimbursement (9 Euro per hour). In addition, participants received a performance-dependent bonus of up to 8 Euro (see Procedure and experimental task). Study approval was obtained by the ethical board of the Department of Psychology at the Ludwig-Maximilians-University Munich and all participants provided written consent at the beginning of the experimental session.

2.2. Apparatus and measurement setup

EEG recording employed 64 active electrodes, positioned according to an extended version of the international 10–20 system (actiCAP snap, Brain Products, Munich, Germany). Electrode FCz was used as an online reference and one additional ground electrode was placed at location Fpz. Data was recorded using BrainAmp DC amplifier (BrainVision) and digitized with a sampling rate of 500 Hz. An online bandpass filter with half-amplitude cutoffs of 0.016 Hz and 250 Hz was applied. Electrode-to-skin impedances were kept below 20 kΩ throughout the experiment.

2.3. Procedure and experimental task

The experimental task was written in MATLAB R2020B, using Psychophysics Toolbox extensions (Psychtoolbox Version 3; Brainard & Vision, 1997). Participants were seated approximately 90 cm in front of a 24-inch monitor on which the task was presented. To study changes in affective feedback processing as a function of sense of agency, we implemented a reinforcement learning paradigm, varying the degree of choice within participants (free-choice/forced-choice) across blocks (Fig. 1). All stimuli were presented on a grey background, with a visual angle of 1.2°.

At the start of each trial, participants had to fixate the white cross presented at the center of the screen (1.5 s). Subsequently, two items in the form of differently colored squares appeared horizontally aligned adjacent to the center (1.0 s). During free-choice blocks, participants were asked to choose one of the two items by pressing the left- or the right-arrow key with their right index finger. During forced-choice blocks, the computer pre-selected one item, indicated by a white rectangle around one of the squares, and participants' only option was to confirm the computer's choice with the corresponding button press. Following keystrokes, participant choices were confirmed with a black rectangle around the selected stimulus for 1.0 s. Next, participants were presented with feedback for item selections (1.0 s). For each block, one of the two colored squares was randomly allocated to be the high-value item, while the other one was allocated to be the low-value item. High-value item selections resulted in a gain of 4 cents (positive feedback) in 75% of all trials. The selection of the low-value item resulted in a loss of 2 cents (negative feedback) in 75% of all trials. Employing gains twice as large as losses was based on research showing that humans value losses approximately twice as much as gains (Proudfit, 2015; Tversky & Kahneman, 1992). Participants were naïve to the allocated action-effect probabilities at the start of each block. To maximize gains and minimize losses, they thus had to use the feedback to learn the item-outcome associations within each block. The implemented probabilities allowed participants to build stable outcome predictions, while still encouraging the exploration of all response options.

Importantly, participants could only freely choose between the items during free-choice blocks (high agency), but not during forced-choice blocks (low agency). If participants failed to give a response or selected the item not chosen by the computer, they were told to 'Respond Faster!' or that the 'Wrong Key [was] Pressed!'. To warrant motivational salience of action effects, participants received the reward they earned during their two most successful free- and forced-choice blocks in the end of the experiment. Depending on participants' (non-) response, trial durations varied between 3.5 s and 4.5 s. Participants completed a total of 640 trials, split into eight blocks, consisting of 80 trials each. There were four blocks for both the free-choice and the forced-choice condition, resulting in 320 trials per agency condition. A switch between free- and forced-choice blocks occurred after every second block. All participants started with two free-choice blocks. Unbeknown to participants, the trial-wise location of the high-value and low-value item as well as the computer choices during forced-choice trials exactly mimicked those of the free-choice trials from the two preceding blocks, thereby resulting in the same order and frequency of high-value and low-value item selections in free- and forced-choice blocks. Participants' confirmation of externally determined item choices ensured that they paid attention to item selections made in forced-choice blocks and that they performed the same motor actions in both agency conditions. Consequently, the only aspect differing between the two conditions was whether participants were endowed with the freedom to choose or not.

The colors of the items were randomly selected from eight previously

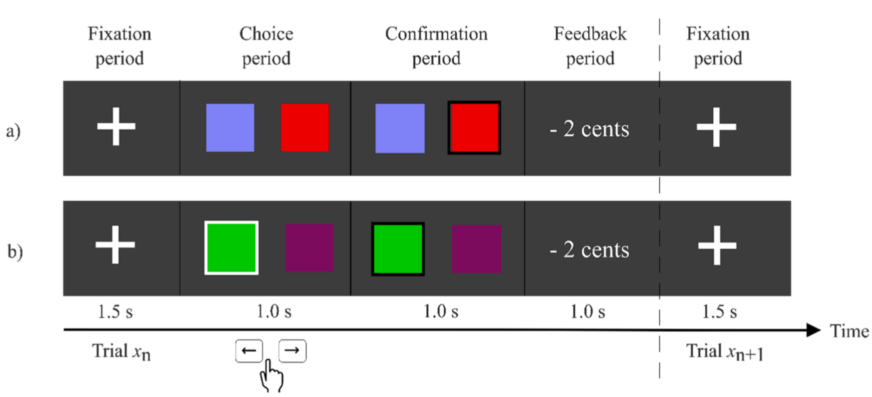

Fig. 1. Schematic task overview. After fixation, participants either a) chose one of the two items presented on screen (free-choice), or b) selected the item chosen by the computer (forced-choice), indicated by a white rectangle around the stimulus. Choices were confirmed with a black rectangle around the selected item. Probabilistic affective feedback was presented as action effect. Free- and forced-choice trials varied across blocks. Item colors and associated gain and loss probabilities remained the same within but changed between blocks.

generated stimulus pairs, such that each stimulus pair was used in exactly one block per participant. To ensure good stimulus discriminability, and to prevent illusory carry-over effects from learning in previous blocks, all stimuli were created to be maximally perceptually different to each other and the background by reference to the CIELAB color space (Holy, 2022). Within each block, both items occurred equally often on the left and right side of the screen in a randomized order. In addition, choosing the high- or low-value item resulted in the corresponding positive or negative feedback in 75% of the trials on each side.

At the end of each block, participants were informed about their monetary gain in the current block and the choice context of the following block. In addition, participants were asked to indicate their agency experience in a paper-pencil format. Participants reported their agreement with two statements (statement 1: 'I was free to choose any item I wanted this block.'; statement 2: 'I was in control over the amount of money I gained in this block.') on a bipolar quasi-continuous scale (Chimi & Russell, 2009), ranging from *Strongly Disagree* to *Strongly Agree*. The items were adapted from previous studies using explicit agency measures (Beyer et al., 2017; Dewey & Carr, 2013). While statement 1 asked participants to state in how far they could influence item choices (choice agency), statement 2 asked them to indicate the degree to which they had control over the feedback they received (outcome agency). Assessing outcome agency in addition to choice agency owed to findings from prior research, showing that the extent of self-determined action influences an agents' perceived control over action outcomes (Barlas et al., 2018). Statement 2 also served to control for participants' potential insight into the pattern of computer choices during forced-choice blocks. In case participants noticed that computer selections mimicked their choices from previous free-choice blocks, and were therefore always contingent on participant behavior, they would be expected to experience similar outcome agency during free- and forced-choice blocks. After filling out the questionnaire, participants could proceed to the next block in a self-paced manner.

The experimental session took approximately 1 h and 45 min per participant. To allow familiarization with the task, all participants completed 10 free-choice and 10 forced-choice practice trials in the beginning of the experiment. To prevent insight into the implemented reward probabilities and the identical sequence of free-choice and forced-choice trials, item locations, computer choices, and feedback valence were determined randomly during practice trials.

2.4. Data analysis

2.4.1. Behavioral analysis

To assess learning progress within free-choice blocks, we categorized each trial as correct (i.e., high-value item choice) or incorrect (i.e., low-value item choice) and summarized performance as the cumulative proportion of correct actions over bins of five trials. We used a paired samples t-test (two-sided) to evaluate whether participants successfully acquired the implemented item-outcome associations, testing whether the proportion of participants' high-value item choices was significantly larger than the proportion of low-value item choices across all free-choice blocks.

The experimental task was designed to only differ in the degree of choice across conditions. However, it is important to consider that free-compared to forced-choice trials may differ in reaction times. These variations in motor activation timing could potentially influence neuronal activity during feedback presentation in the progression of the trial. In addition, a high occurrence of non-responses in either the free- or forced-choice blocks could unintentionally introduce a disparity in the frequency of positive and negative feedback between the different agency conditions. We therefore tested for differences in reaction times and the proportion of positive and negative feedback between free- and forced-choice blocks using paired samples t-tests (two-sided).

To evaluate whether our manipulation was successful in inducing either high or low levels of sense of agency and to assess whether participants' feeling of control over action outcomes varied between free- and forced-choice blocks, a repeated measures analysis of variance (ANOVA) with the factors CHOICE (free-choice/forced-choice) and RATING (choice agency/outcome agency) was performed on the questionnaire responses. To this end, participants' agency ratings were quantified as percentages, computed as the degree of agreement with statement 1 and statement 2 for free- and forced-choice blocks separately (Wegner & Wheatley, 1999). Greenhouse Geiser corrections were applied if sphericity was violated. Interaction effects were subsequently examined using paired-samples t-tests (two-sided), and test statistics corrected for multiple comparisons using Holm's method (1979).

Statistical tests employed an alpha level of .05 and effect sizes for all tests are reported using Cohen's d (small effect: $d = 0.20$; medium effect: $d = 0.50$; large effect: $d = 0.80$) or partial-eta-squared (small effect: η_p^2 =.01; medium effect: η_p^2 =.06; large effect: η_p^2 =.14; Cohen, 1988). We additionally examined the data by estimating Bayes Factors (*BF*) for individual tests using Bayesian Information Criteria (Wagenmakers, 2007). *BF*s quantify the evidence for the null hypothesis (H_0) as well as the alternative hypothesis (H_1), and allow to evaluate the likelihood of observing the alternative hypothesis ($\mu_1 \neq \mu_2$) compared to the null hypothesis ($\mu_1 = \mu_2$; weak evidence for H_1: BF_{10} = 1–3; positive evidence for H_1: BF_{10} = 3–20; strong evidence for H_1: BF_{10} = 20–150; very strong evidence for H_1: $BF_{10} > 150$; Raftery, 1995). Behavioral analyses were performed in MATLAB R2020B and RStudio.

2.4.2. Time-frequency analysis

2.4.2.1. Data preprocessing. We performed data pre-processing offline within MATLAB R2020B, using custom-written scripts and Fieldtrip toolbox (Oostenveld et al., 2011). The EEG signal was filtered using a Butterworth Infinite Impulse Response two-pass filter (high-pass: 1.0 Hz; low-pass: 40 Hz; roll-off: 6 dB/octave) and re-referenced to an average of all active electrodes (see Kaiser et al., 2021). Trial-wise epochs of 4500 ms, from 2000 ms prior to 2500 ms post feedback onset, were extracted. To save processing time, the data were down-sampled to 250 Hz. Artifacts due to eyeblinks and eye movements were identified using independent component analysis (using the runica algorithm implemented in Fieldtrip) and removed, resulting in the exclusion of one to four components ($M = 2.6$ components, $SD = 0.8$) per participant. The remaining components were projected back to the data at channel level. To identify trials with substantial noise, voltage deflections exceeding ± 100 μV were identified in the baseline corrected dataset, using 200 ms prior to feedback onset as baseline. Noisy trials were subsequently removed from the non-baseline corrected data, resulting in a mean removal of 5.3% (SD = 3.8, range: 0.2–16.7) of all trials. Further, one to two exceedingly noisy electrodes were removed for four participants ($M = 1.3$ electrodes, $SD = 0.5$) and replaced with spherical spline interpolation using the Fieldtrip function *ft_channelrepair* (see also Perrin et al., 1989). Next to removing noisy data, trials for which participants did not receive feedback (either because they did not respond within the 1000 ms time window of stimulus presentation, or because they chose the item not indicated by the computer during forced-choice blocks) were excluded from further analyses, resulting in an average removal of 16.4 trials (SD = 17.1, range: 1–72).

Successful reinforcement learning implies a gradual decrease in low-value item choices within each block. This pattern was also reflected in participants' behavior in the current study (see Behavioral results). As expected, over the course of each block, participants selected high-value items most of the time, leaving only an exceedingly low number of trials with low-value item choices. We thus focused our main statistical analysis on trials with high-value item choices, resulting in an average exclusion of 83.9 trials (SD = 67.1, range: 12–288). Finally, participants were excluded from subsequent analyses, if less than 50 trials were retained for the main analysis in at least one condition after preprocessing (a priori criterion, see Kaiser & Schütz-Bosbach, 2021), leading to an exclusion of two participants. The final dataset comprised of 30 participants with an average of 521.4 trials (SD = 50.6, range: 436–602). Condition wise averages are depicted in Table 1.

To increase topographical specificity, a Surface Laplacian, using the spherical spline method (Perrin et al., 1989) with a polynomial degree of 10 (Cohen, 2014), was applied to the pre-processed data (*ft_scalpcurrentdensity*). Condition-wise averages for the time-frequency data depicting midfrontal oscillatory activity were calculated over a time window from –500 ms to2000 ms relative to feedback onset. This time window exceeded the time period of interest to avoid edge effects (Cohen, 2014). For precise localization of frequency information in time, Morlet wavelet convolution was applied, with cycles and frequency linearly increasing from three to eight, and 2 Hz to20 Hz, respectively (Cohen, 2014; Kaiser & Schütz-Bosbach, 2021). Each sample from the time-frequency epochs of all electrodes was baseline corrected using the average power over all trials and converted to the decibel (dB) scale (dB = 10 *log10[power/baseline]), with a baseline window ranging from −300 ms to −100 ms relative to feedback onset (Kaiser et al., 2022).

2.4.2.2. Statistical analysis. The primary goal of our analysis was to assess changes in the neuronal processing of affective feedback as a function of sense of agency. Theta oscillations associated with feedback processing have frequently been reported to be most prominent over FCz and adjacent electrodes (e.g., Hajihosseini & Holroyd, 2013; Kaiser et al., 2021; Luft et al., 2013). We thus created time-frequency maps and performed statistical analyses using the averaged activity over electrodes Fz, FCz, FC1, and FC2. Topographical plots, depicting average theta power from 200 ms to 600 ms after feedback onset (Cohen et al., 2007; Luft et al., 2013), confirmed that activity was strongest around these scalp sites, especially for free-choice trials with negative feedback (Fig. 2).

We used cluster-based permutation analysis over the specified scalp sites to assess the main effects of choice, feedback valence, and their interaction (Maris & Oostenveld, 2007) during feedback presentation. Univariate repeated measures ANOVAs (*ft_statfun_depsamplesFunivariate*) were performed to quantify the main effects for the factors CHOICE (free-choice/forced-choice) and VALENCE (positive/negative) at the sample level. To examine whether the neural impact of choice depended on feedback valence, contrasts for a CHOICE x VALENCE interaction were calculated by subtracting mean power values evoked by negative feedback from mean power values evoked by positive feedback within each choice condition. Both conditions were subsequently entered as factors in the univariate ANOVA and sample significance determined identically to the main effect analyses. To examine the distinct effect of choice for both positive and negative feedback, we performed additional contrasts of free- and forced-choice trials for each level of feedback valence separately via two-sided cluster-based permutation t-tests (*ft_statfun_depsamplesT*).

Significance of individual data points was evaluated against at an alpha level of .05, and samples passing the significance threshold clustered based on temporal and spectral features. We subsequently computed cluster-level statistics, defined as the sum of the F-values (or t-values, respectively) within each cluster. To evaluate significant differences across conditions, we applied the Monte Carlo Method to the largest of these clusters using 1000 random permutations of the original data. This method calculates p-values under the permutation distribution for data consisting of temporally and spatially adjacent data points by estimating the probability with which the condition-wise cluster data come from the same, or different distributions, while simultaneously controlling for the false alarm rate (Maris & Oostenveld, 2007).

Table 1
Mean Number of Trials (and Standard Deviations) for Each Combination of Trial Type (Free-Choice, Forced-Choice) and Feedback Valence (Positive, Negative) in the Final Dataset.

Trial type	Feedback valence	
	Negative	Positive
Free-choice	66.5 trials (6.4)	196.9 trials (19.5)
Forced-choice	65.3 trials (6.3)	192.9 trials (20.4)

3. Results

3.1. Behavioral results

Averaged over entire blocks, participants chose the high-value item significantly more often (M = 88.1%, SD = 7.2, range: 226–312) than the low-value item (M = 11.9%, SD = 7.2, range: 5–83), indicating that they were successful in learning the implemented item-outcome associations, $t(29) = 28.29$, $p < .001$, CI 95% [221.6, 256.1], Cohen's $d = 10.26$, $BF_{10} > 10^6$ (Fig. 3).

Reaction times had a mean latency of 524.80 ms (SD = 58.56, range: 424.41–681.55) and did not significantly differ between free- and forced-choice trials, M_{diff} = 2.88 ms, $t(29) = 0.63$, $p = .54$, CI 95% [− 6.53, 12.28], Cohen's $d = 0.05$, $BF_{10} = 0.23$. Participants received positive feedback on an average of 68.5% (SD = 3.9) trials and earned a final monetary bonus of 7.20 Euro (SD = 0.62, range: 5.64–7.92). No significant differences between free- and forced-choice blocks could be found for the proportion of positive feedback (M_{diff} = 1.7 trials, $t(29) = 1.48$, $p = .15$, CI 95% [− 0.64, 4.04], Cohen's $d = 0.12$, $BF_{10} = 0.52$), or negative feedback, $M_{diff} = -0.37$ trials, $t(29) = -0.50$, $p = .62$, CI 95% [− 1.87, 1.13], Cohen's $d = -0.03$, $BF_{10} = 0.22$. To conclude, reaction times and the rate of positive and negative feedback did not differ between free-choice and forced-choice blocks.

Inspection of participants' agency ratings for choice and outcome agency measures revealed five extreme outliers (defined as values < [1st quartile – 3 *interquartile range] or > [3rd quartile + 3 * interquartile range]). Reported results are based on the dataset including outliers, as their exclusion did not affect the results. The repeated measures ANOVA revealed a large effect of CHOICE on agency experience, $F(1, 29) = 169.16$, $p < .001$, $\eta_p^2 = .854$, $BF_{10} > 10^6$. Participants experienced greater agency during free- (M = 69.76%, SD = 23.37) compared to forced-choice blocks (M = 9.52%, SD = 16.90). In addition, we found evidence for a significant effect of RATING on agency experience, $F(1, 29) = 16.60$, $p < .001$, $\eta_p^2 = .364$, $BF_{10} = 0.69$. Participants reported greater levels of choice agency (M = 45.13%, SD = 40.97) than outcome agency (M = 34.15%, SD = 30.62) across blocks. The CHOICE x RATING interaction was also significant, indicating that the effect of choice differed for the two agency ratings, $F(1, 29) = 14.21$, $p < .001$, $\eta_p^2 = .329$, $BF_{10} = 28.70$. Post-hoc analyses revealed a larger effect of CHOICE on choice agency ratings (i.e., 'I was free to choose any item I wanted in this block.'), $t(29) = 12.04$, CI 95% [0.58, 0.82], $p < .001$, Cohen's $d = 3.41$, $BF_{10} > 10^6$, than on outcome agency ratings (i.e., 'I was in control over the amount of money I gained in this block.'), $t(29) = 10.40$, $p < .001$, Cohen's $d = 2.88$, $BF_{10} > 10^6$ (Fig. 4). The results indicate that our manipulation was successful in influencing agency experience between free- and forced-choice blocks. The larger effect of CHOICE on choice agency than outcome agency ratings additionally shows that participants were sensitive to their variable influence on item choice and action outcome, wherein self-determined choices did not always result in the desired outcome (i.e., a monetary gain).

3.2. Midfrontal oscillatory activity

Fig. 5 depicts time-frequency maps for midfrontal oscillatory power relative to feedback onset. Results of cluster-based permutation analyses for the main effects of choice, feedback valence, as well as their interaction are shown in Fig. 6.

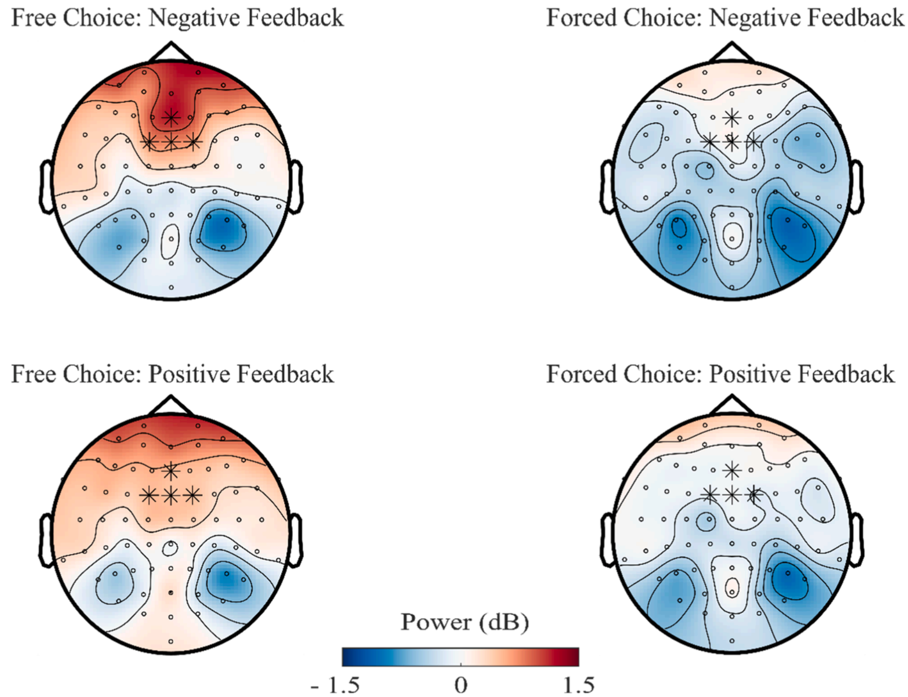

Fig. 2. Topographical plots showing oscillatory activity in the theta range (4–7 Hz) 200 600 ms post feedback onset. Stars mark electrodes used for statistical analyses.

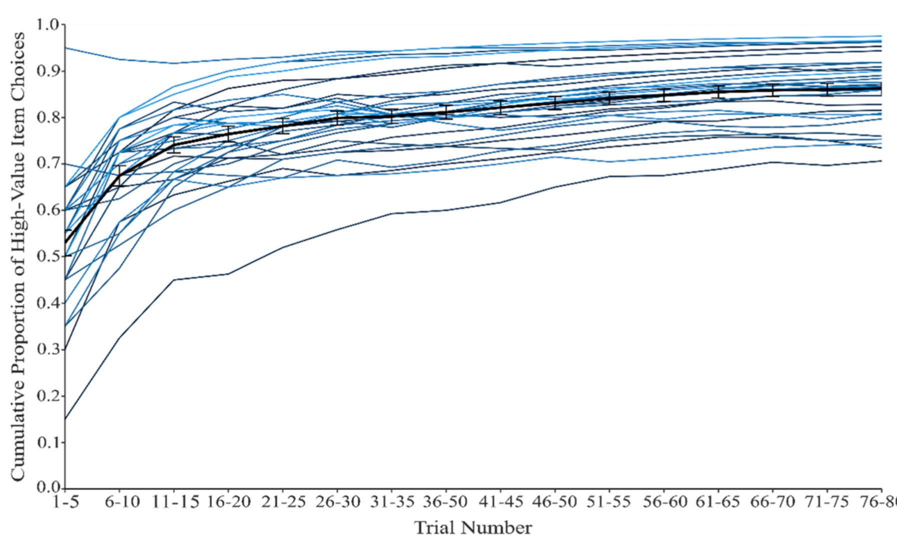

Fig. 3. Cumulative proportion of high-value item choices across trials. Blue-shaded lines indicate learning curves of individual participants, averaged over all free-choice blocks. The black line depicts the grand average of choice behavior across all participants, including standard errors.

Our analyses revealed a significant effect of CHOICE on feedback processing over midfrontal electrodes, encompassing low frequency oscillations mainly in the theta range and starting with feedback onset, 0–930 ms, $p = .001$. Free-choice trials elicited greater theta power during feedback presentation than forced-choice trials. An average power difference of 0.69 dB ($SD = 0.22$) could be observed for the detected cluster. The ANOVA for the main effect of VALENCE revealed a significant power difference between trials with negative and positive feedback, confined to the theta range and starting with feedback onset, 0–510 ms, $p = .022$. Compared to trials with positive feedback, trials with negative feedback elicited greater theta power ($M\Delta = .29$ dB, $SD = 0.09$) during feedback presentation. We did not find a significant effect for the interaction of the factors CHOICE and VALENCE during the specified time window. This indicates that changes in the sense of agency affect the processing of positive and negative feedback, as observable in midfrontal theta activity, in comparable ways. Additional contrasts confirmed the significant difference between free and forced choices for both positive and negative feedback (Fig. 7). Free- compared to forced-choice trials significantly increased activity in the theta range after the onset of negative feedback, 0–830 ms, $M\Delta = .85$ dB, $SD = 0.24$, $p = .001$. For positive feedback, a significant increase in theta power could be observed for free- compared to forced-choice trials for the whole duration of feedback presentation, 0–1000 ms, $M\Delta = .60$ dB, $SD = 0.16$, $p = .004$.

4. Discussion

The current study investigated the impact of changes in the sense of

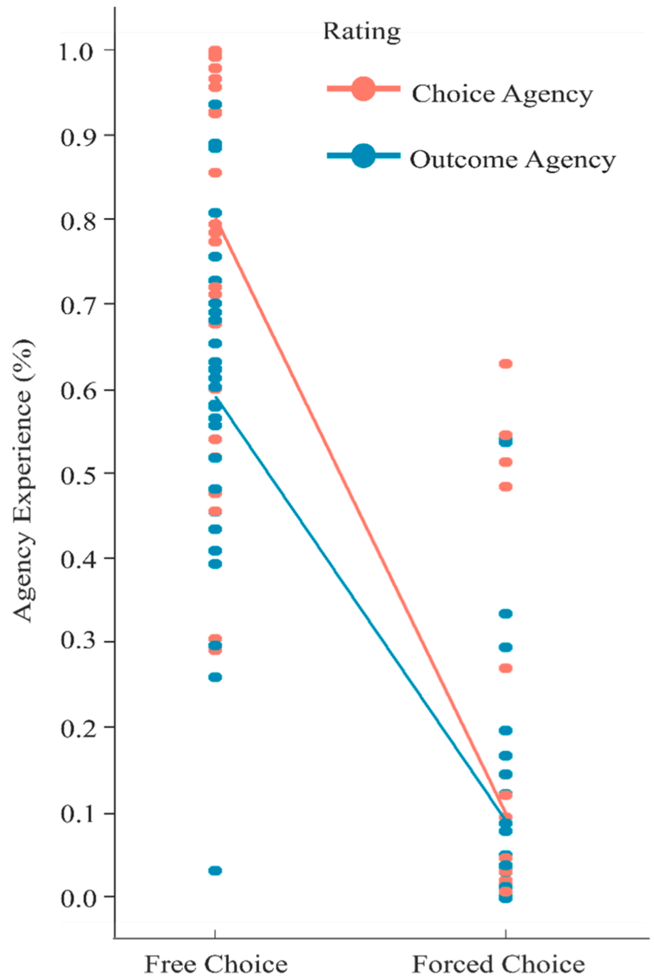

Fig. 4. Participants' proportional agreement with statements measuring choice agency (i.e., 'I was free to choose any item I wanted in this block.') and outcome agency (i.e., 'I was in control over the amount of money I gained in this block.'), plotted separately for free- and forced-choice blocks. Lines depict the mean difference in agency experience between free and forced choices for both choice (red) and outcome agency ratings (blue).

agency on the neuronal processing of affective action outcomes. This was achieved by manipulating freedom of choice within subjects in a reinforcement learning task, with monetary gains and losses as positive and negative feedback. Using EEG, we analyzed midfrontal oscillatory activity during feedback presentation and assessed differences between free and forced choices, as well as between positive and negative feedback. Explicit agency ratings, measuring perceived freedom of choice and outcome controllability, were obtained after each free- and forced-choice block.

Behavioral results show that participants quickly gained insight into the reward structure, with an increase in high-value item choices throughout free-choice blocks. Reported sense of agency was significantly higher during free- than forced-choice blocks, showing that our manipulation was effective in modifying agency experience during the task. The effect of choice was larger for perceived freedom of choice than the feeling of control over monetary gains. As hypothesized, negative compared to positive feedback elicited greater theta power in the region of interest. Also, we found an increase in midfrontal oscillatory activity in the theta range for free compared to forced choices during feedback presentation. Notably, our results reveal that freedom of choice amplifies the neuronal impact of feedback during the processing of both positive and negative action outcomes.

The present study substantiates the evidence on agency-related changes in the processing of affective action outcomes. Increased midfrontal theta power for negative compared to positive feedback, as well as for self-, rather than externally determined action outcomes mirrors results from previous studies focusing on the FRN (e.g., Bellebaum et al., 2010; Foti et al., 2015; Hassall et al., 2019; Weismüller et al., 2019; Yeung et al., 2005). By analyzing time-frequency oscillations, our findings complement previous ERP work and provide insight into the oscillatory dynamics underlying feedback processing during goal-directed tasks (Cavanagh et al., 2010; Luft, 2014).

Our paradigm was designed to manipulate freedom of choice, while maintaining a consistent level of personal motor involvement and an equal rate of positive and negative feedback across all conditions. To this end, participants performed motor actions in free- and forced-choice blocks and item selections and reward schemes were kept identical for both choice conditions. Importantly, behavioral analyses revealed that reaction times did not significantly differ between free and forced choices. This indicates that the observed differences in neural reactivity to feedback following participants' responses were due to differences in freedom of choice rather than discrepancies in the timing of motor reactions between conditions. Our results thus allow to draw conclusions about the distinct contribution of choice on affective feedback processing during goal-directed actions.

As suggested by theoretical accounts (Friston et al., 2013; Ly et al., 2019) and empirical evidence from previous studies (Barlas & Obhi, 2013; Haggard, 2017; Hassall et al., 2019), freedom of choice increases agency experience. By employing explicit measures that ask participants to report their perceived control over action outcomes, most previous studies have treated the sense of agency as a single construct. However, in our study, we utilized a self-report measure that explicitly differentiated between the perceived freedom of action choice (i.e., 'I was free to choose any item I wanted in this block.') and the ability to influence the outcome of that action (i.e., 'I was in control over the amount of money I gained in this block.'). It is important to note that having the freedom to make decisions does not guarantee the desired outcome, underscoring the importance of distinguishing between these components when assessing the sense of agency.

As apparent from agency ratings in the current study, freedom of choice is closely linked to the sense of control over the outcome. The decrease in perceived outcome controllability during forced-choice blocks is in line with previous work (Barlas et al., 2018). In addition, due to the substantial number of trials within each block and the inclusion of a non-corresponding block between two corresponding free-choice and forced-choice blocks, there was always a time gap of approximately 10 min between the start of two corresponding blocks. Considering our behavioral findings and the task structure, it is highly unlikely that participants were able to discern the match of their own choices and those made by the computer.

Importantly, the larger increase in choice than outcome agency ratings for free compared to forced choices shows that participants are sensitive to their variable influence on distinct task components. In the current study, items and outcomes were associated probabilistically, implying that outcomes could not always be predicted from item choices. A smaller increase in outcome than choice agency ratings during free choices thus shows that participants correctly determined that increased freedom of choice did not always guarantee positive feedback, illustrating the influence of various task elements on an individual's sense of agency.

Higher feedback-evoked theta activity for free-choice than forced-choice trials converges with insights from prior work (e.g., Weismüller et al., 2019; Zheng et al., 2020). Crucially, our study reveals that the augmented midfrontal theta power observed during feedback presentation following free choices extends to contexts of significant motivational value. By providing participants with a reinforcement learning task wherein they could maximize a performance-dependent bonus paid to them at the end of the experiment, we combined two aspects relevant to the motivational salience of actions and their outcomes (Ly et al.,

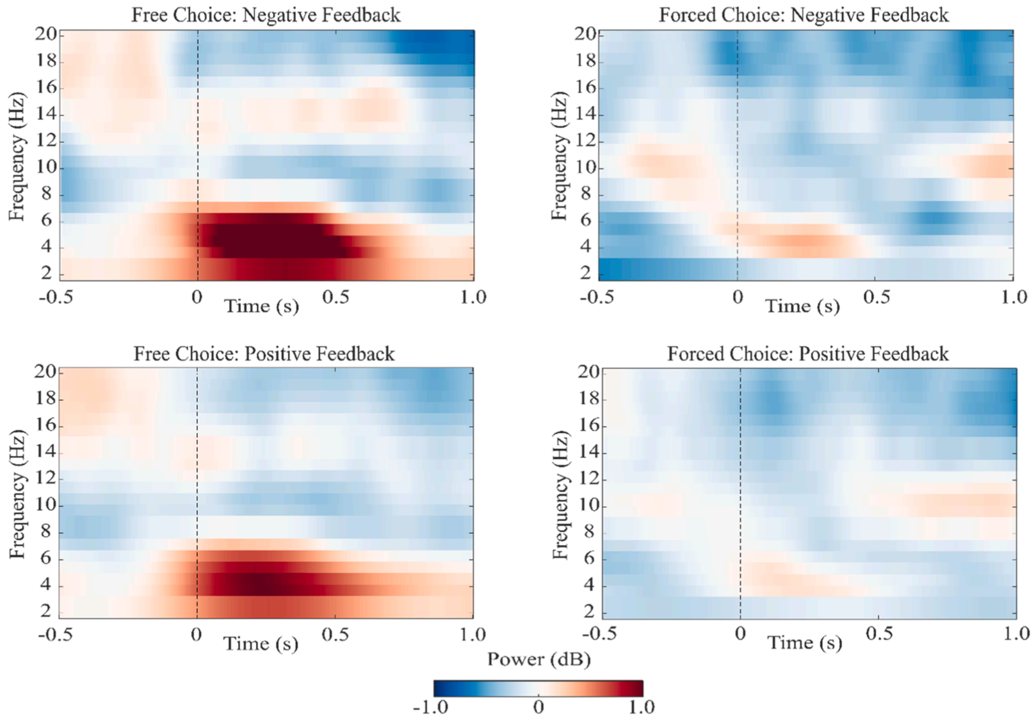

Fig. 5. Time-Frequency plots over midfrontal electrodes (Fz, FCz, FC1, FC2) for all choice-outcome conditions. Dashed line at Time = 0 marks the onset of feedback presentation.

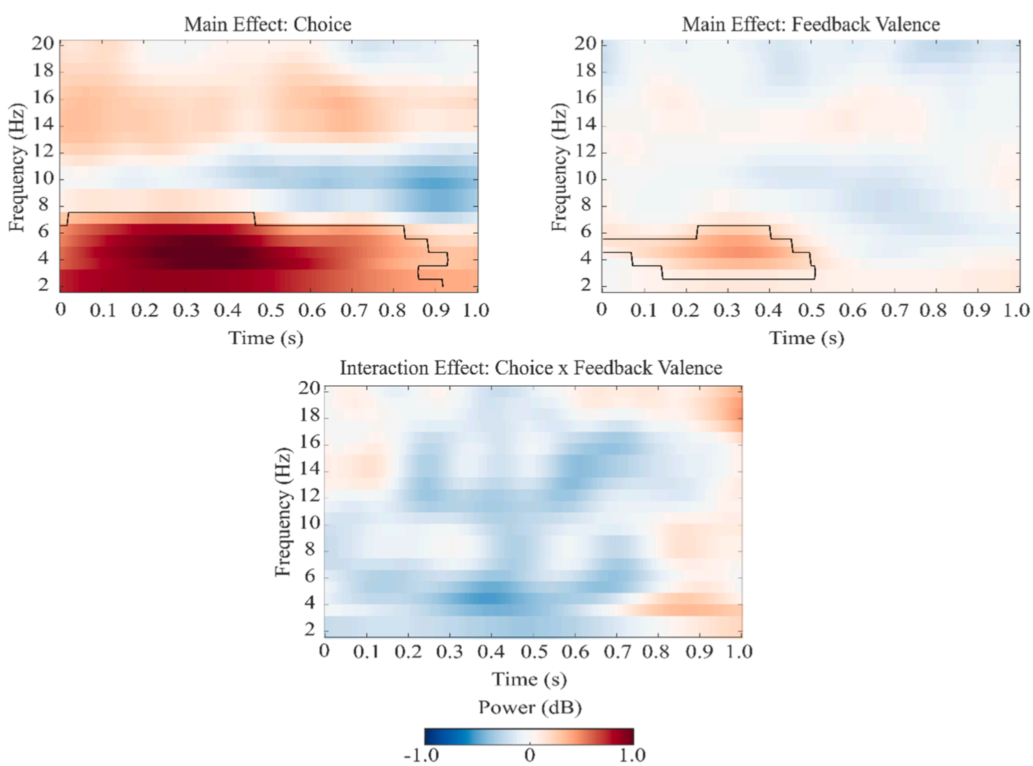

Fig. 6. Main and interaction effects of choice and feedback valence, averaged over midfrontal electrodes (Fz, FCz, FC1, FC2). Time = 0 indicates the onset of feedback presentation. Black contours mark the time and frequency range with significant differences between the examined conditions ($p < .05$).

2019; Mei et al., 2018). Sense of agency has been proposed to be particularly relevant in such contexts, allowing for effective regulation of behavior to attain desirable and avoid undesirable states (Kaiser et al., 2021; Ly et al., 2019; Murayama et al., 2015). Accounting for the goal-directedness of behavior therefore contributes to an understanding of how the sense of agency affects the processing of action outcomes in situations central to its functional role.

Midfrontal theta oscillations are suggested to play an important role for performance monitoring and the updating of task-relevant information (e.g., Cavanagh et al., 2010; Kaiser et al., 2021; Luft, 2014; Luft

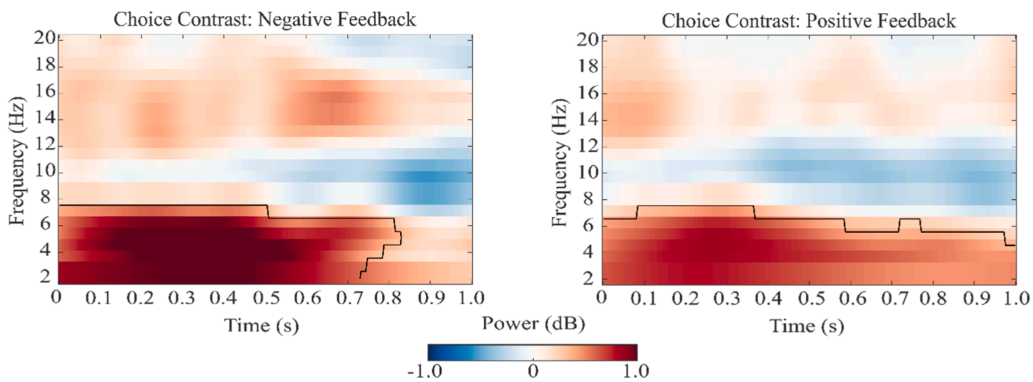

Fig. 7. Results of contrast analyses testing the effect of choice for negative (left) and positive (right) feedback, averaged over midfrontal electrodes (Fz, FCz, FC1, FC2). Time = 0 indicates the onset of feedback presentation. Black contours mark the time and frequency range with significant differences between the examined conditions ($p < .05$).

et al., 2013). With respect to these insights, the results from the present study might indicate that participants track affective feedback obtained for externally determined choices to a lesser extent than feedback obtained for self-determined choices, reflected in lower midfrontal theta power during forced- compared to free-choice trials. This choice-related decrease in theta power might reflect participants' perception of reduced personal responsibility for gains and losses resulting from externally determined actions. In line with this interpretation, diminished sense of agency has been associated with reduced outcome monitoring (Beyer et al., 2017). In addition, the motivation to learn from performance feedback might be decreased if it cannot be used to adjust behavior (Moscarello & Hartley, 2017). Both aspects signal a decrease in personal task involvement during forced choices, potentially explaining the reduced impact of affective feedback during task performance. It should be noted, however, that while our results clearly show that freedom of choice increases the neural impact of affective feedback, our study design does not allow to draw conclusions about potential behavioral and cognitive effects of agency-related changes in affective feedback processing.

Heightened midfrontal theta power for negative compared to positive feedback resonates with a large body of work on affective feedback processing in the context of reinforcement learning (see Luft, 2014). In the current task, action outcomes always affected participants' payout, meaning that feedback carried personal relevance during both free- and forced-choice blocks. Accordingly, our results suggest that midfrontal theta oscillations differentiate between positive and negative feedback for all self-relevant action outcomes, regardless of whether they result from self- or externally determined choices. Thus, whilst the main effect of choice indicates that being able to freely decide between response options increases personal task involvement, the mere presentation of adverse self-relevant feedback seems to be sufficient to elicit a surge in midfrontal theta power.

Notably, free compared to forced choices not only increased the neural impact of negative feedback, but also intensified the processing of positive feedback. While previous studies led to inconsistent conclusions regarding the question if agency selectively enhances the processing of positive or negative feedback (e.g., Kaiser et al., 2021; Weismüller et al., 2019; Zheng et al., 2020), our results clearly demonstrate midfrontal theta activity as a neurophysiological marker sensitive to within-subject differences in agency experience for both types of affective feedback. Thus, rather than exhibiting a positivity or negativity bias, the valence-independent increase in oscillatory power during free choices indicates midfrontal theta oscillations as a general mechanism for outcome monitoring during goal-directed actions.

The valence-independent increase in midfrontal theta power for feedback following free choices is inconsistent with Zheng et al. (2020), who found that the effect of choice was larger for negative than positive feedback. The origin of this divergence conceivably lies in the nature of the implemented tasks. Whereas Zheng et al. employed a pre-defined reward scheme with equiprobable gain and loss feedback, which did not allow participants to learn from feedback to optimize their behavior, the current study used a reinforcement learning task with informative, motivationally salient feedback. While free choices seem to increase the relevance of both positive and negative feedback in a reinforcement learning context, the processing of negative feedback appears to take precedence in non-learnable contexts. This indicates that the evaluation of trial-wise gains and losses for free and forced choices scales differently across affective contexts. In non-learnable environments, a selective increase in midfrontal theta power for negative feedback might reflect negative affect associated with the discrepancy between desired and actual state (Botvinick et al., 2001; Cavanagh et al., 2010). When agents are presented with learnable environments, however, performance monitoring becomes increasingly salient (Moscarello & Hartley, 2017), which might be reflected in a general enhancement of midfrontal theta activity during feedback presentation for free choices. Hence, by actively tracking both positive and negative action outcomes, agents not only ascertain what behavior to inhibit, but also learn to repeat rewarded actions.

Some limitations should be kept in mind when interpreting the current results. As all participants started with two free-choice blocks, we cannot exclude that order effects influenced the present results. To match reward rates between free and forced choices, while maintaining a certain degree of outcome controllability in free-choice trials, however, presenting free-, prior to forced-choice blocks was inevitable. Furthermore, as we provided participants with a relatively simple reinforcement learning task without explicit learning requirements in forced-choice blocks, the current study design was not suited for examining free-choice and forced-choice differences in learning per se. Thus, it is important to acknowledge that there are at least two potential explanations for the differences in feedback processing between free and forced choices in the current study. Firstly, the differences in theta power during feedback presentation might reflect variations in learning for self- compared to externally determined action outcomes. Secondly, participants may experience a heightened sense of personal responsibility for the consequences of their own choices compared to externally determined choices, which may not necessarily be linked to any learning effect. Future research is needed to disentangle these possibilities. In case participants exhibit agency-related differences in learning, midfrontal theta should be tested as a neuronal mechanism explaining this relation. When examining midfrontal theta activity as a mediator of agency and action regulation, it might additionally be useful to distinguish between effects pertaining to perceived freedom of choice on the one, and outcome controllability on the other hand. As agency ratings from the current study indicate, participants were able to differentiate between their ability to choose their action and their ability to influence the result of this action (i.e., whether they receive a monetary gain or loss). By investigating the influence of individual cognitive and motor factors on the neuronal processing of affective action outcomes, we can establish a valuable groundwork for comprehending the inconsistent findings observed in prior studies regarding the effects of agency on affective feedback processing (Kaiser et al., 2021).

Together, the current findings implicate midfrontal theta power as a neurophysiological marker sensitive to freedom of choice in a reinforcement learning task. Importantly, while previous studies found inconsistent results concerning the valence specificity of agency-related effects on feedback processing in goal-directed actions, the present study suggests that freedom of choice enhances the processing of both positive and negative performance feedback. Future work should assess the behavioral consequences of these neural effects and explore whether the enhanced processing of affective feedback during learning extends to other neurophysiological markers previously associated with the sense of agency.

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CRediT authorship contribution statement

Maren Giersiepen: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - original draft, Project administration; Simone Schütz-Bosbach: Conceptualization, Writing - review and editing, Supervision, Funding acquisition; Jakob Kaiser: Conceptualization, Methodology, Software, Formal analysis, Writing - review and editing, Supervision, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors did not use generative AI technologies for preparation of this work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data and study materials can be found online at https://osf.io/nc945/?view_only= 94061c5c5e954c20b758cc3d479afe77.

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